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SUMMER FEST
BARCELONA



presenta...

AI-Powered Multilingual Reports with Azure AI and Semantic Link Labs

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isolutions

Samantha Cruz

Senior Data Engineer at isolutions FP
Technical Instructor at Databricks

Also worked for:
Glovo, AB-Inbev, APCOA Parking

Venturing into Microsoft Fabric



Agenda

- Context: When do we need multilingual reports?
- Tools used
- Understanding the Components of a Report
- Limitations

When do we need multilingual reports?

- International companies with global teams
- Regional managers in different countries
- Government and public sector reports
- Customer-facing dashboards
- Legal or compliance reporting
- Multi-brand/multi-country platforms
- Training and onboarding materials
- Accessibility and inclusion initiatives

Tools used: **Semantic Link Labs**

Semantic Link Labs is a Python library designed for use in Microsoft Fabric notebooks. This library extends the capabilities of Microsoft's **Semantic Link** offering additional functionalities to integrate and work with it.

- Export and Modify Semantic Models Programmatically
- Automated Translation of Model Metadata
- Bulk Metadata Editing
- Comparison Between Models
- Manage Role-Level Security
- Generate Documentation Automatically

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semantic-link-labs.readthedocs.io/en/latest/

semantic-link-labs

latest

Search docs

CONTENTS:

sempy_labs



GenAI apps + MongoDB Atlas You don't need a separate database to start building GenAI-powered apps.

Ad by EthicalAds

Welcome to semantic-link-labs's documentation! View page source

Welcome to semantic-link-labs's documentation!

Contents:

- sempy_labs
  - sempy_labs package

Indices and tables

- Index
- Module Index
- Search Page

Next

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Built with Sphinx using a theme provided by Read the Docs.

Tools used: **Azure AI Translator**

Azure AI Translator is a cloud-based service that provides language translation, it uses Neural Machine Translation. Machine translation with REST API call. Supports over 90 languages and dialects.

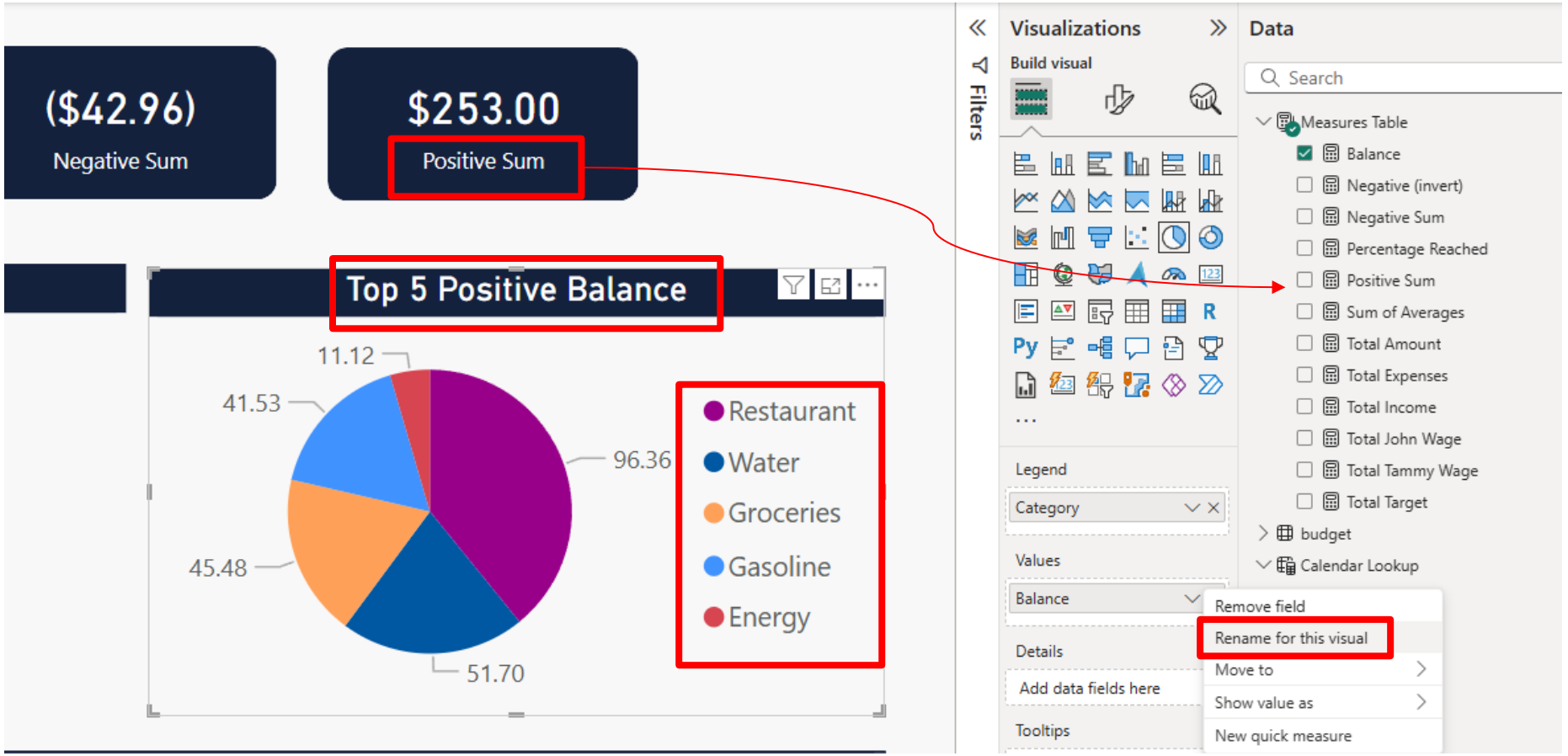
Pricing:

- ➔ Free tier with up to **2 million characters** per month.
- ➔ Pay-as-you-go ranges from €8.85 (standard) to €35.39 (custom) per million characters.
- ➔ Other prices for document translation, training data, custom model hosting, high-volume usage, and annual container options are also available at https://azure.microsoft.com/en-us/pricing/details/cognitive-services/translator/

Understanding the Components of a Report

1. **Data Layer (Content stored in the model)**
   - Table names
   - Column names
   - Measures
   - Data values (text fields)
2. **Metadata Layer (How things are labeled or structured)**
   - Renamed fields
   - Tooltips
   - Field display names
   - Titles of visuals
3. **Report Layer (UI & user-facing content)**
   - Buttons
   - Text boxes
   - Page descriptions
   - Static labels and annotations

Understanding the Components of a Report



https://community.fabric.microsoft.com/t5/Data-Stories-Gallery/Personal-Finance-Dashboard-Painel-de-Finan%C3%A7as-Pessoais/m-p/4671757

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sempy_labs.translate_semantic_model(*dataset: str | UUID, languages: str | List[str], exclude_characters: str | None = None, workspace: str | UUID | None = None*) → DataFrame

Translates names, descriptions, display folders for all objects in a semantic model.

Parameters:

- **dataset** (*str | uuid.UUID*) – Name or ID of the semantic model.
- **languages** (*str, List[str]*) – The language code(s) in which to translate the semantic model.
- **exclude_characters** (*str*) – A string specifying characters which will be replaced by a space in the translation text when sent to the translation service.
- **workspace** (*str | uuid.UUID, default=None*) – The Fabric workspace name or ID. Defaults to None which resolves to the workspace of the attached lakehouse or if no lakehouse attached, resolves to the workspace of the notebook.

Returns: Shows a pandas dataframe which displays all of the translations in the semantic model.

Return type: pandas.DataFrame

Source: https://semantic-link-labs.readthedocs.io/en/stable/sempy_labs.html

Translate a semantic model

```python
import sempy_labs as labs

dataset = '' # Enter the name or ID of your semantic model
workspace = None # Enter the name or ID of the workspace in which the semantic model resides
languages = ['French', 'German', 'Chinese']

labs.translate_semantic_model(dataset=dataset, workspace=workspace, languages=languages)
```

Source: https://semantic-link-labs.readthedocs.io/en/stable/sempy_labs.html

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```python
import sempy_labs as labs

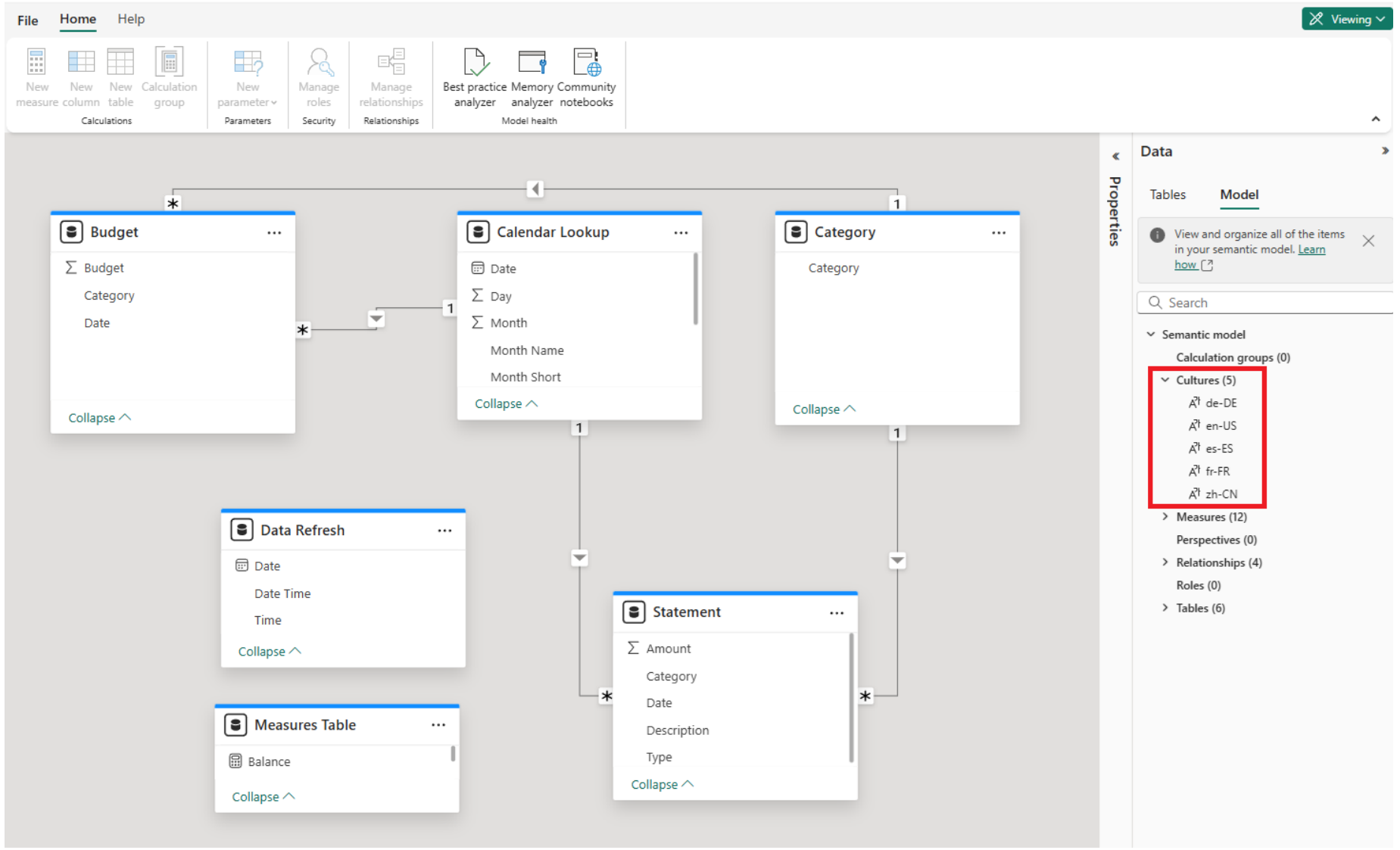
dataset = '79d1f506-3c87-49ee-84c0-95b72aa75743' # Enter the name or ID of your semantic model
workspace = None # Enter the name or ID of the workspace in which the semantic model resides
languages = ['French', 'German', 'Chinese', 'Spanish']

translations_df = labs.translate_semantic_model(dataset=dataset, workspace=workspace, languages=languages)
```

[*] 37 sec - Running PySpark (Python)

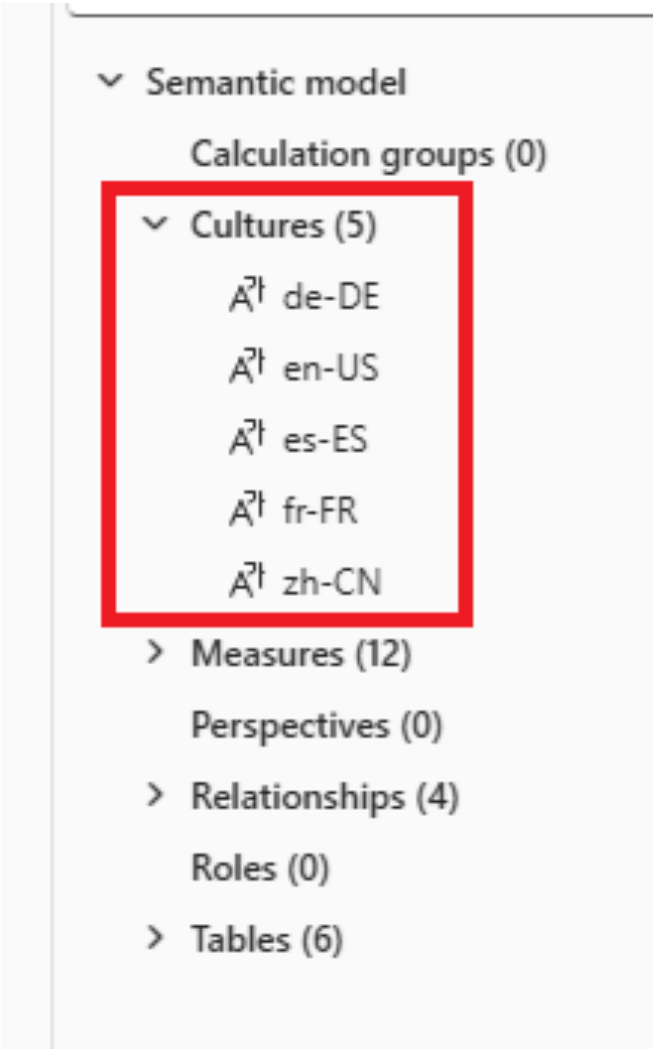
Spark jobs In progress (1) Resources Log

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🧠 Translations become *Cultures* in the Semantic Model

- Each language is stored as a culture (e.g. fr-FR, es-ES, zh-CN)
- Cultures hold translations for table, column, measure metadata
- Power BI auto-selects culture based on the user's language
- Enables multilingual UX with a single report and dataset



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Microsoft Power BI

app.fabric.microsoft.com/reportEmbed?reportId=6b1c46b0-fb8e-4c20-83e7-71b2a71b5acd&autoAuth=true&ctid=31519d70-7496-44fb-9329-e6d5eae4a2da

Monthly Expense Control

Category	Budget	Spent	Balance	% Spent
Miscellaneous	500.00	534.78	-34.78	106.96%
Kids Expenses	350.00	356.31	-6.31	101.80%
Woody the Dog	350.00	351.86	-1.86	100.53%
Car Loan	650.00	650.01	-0.01	100.00%
Apple Cloud	21.50	21.50	0.00	100.00%
Car Insurance	400.00	400.00	0.00	100.00%
Donation	1,700.00	1,700.00	0.00	100.00%
Emergency Funds	850.00	850.00	0.00	100.00%
Internet	150.00	150.00	0.00	100.00%
Investment	1,200.00	1,200.00	0.00	100.00%
Mobile Plan	200.00	200.00	0.00	100.00%
Netflix	30.45	30.45	0.00	100.00%
Office 365	25.60	25.60	0.00	100.00%
Rent	3,200.00	3,200.00	0.00	100.00%
Savings Flexible	500.00	500.00	0.00	100.00%
Sports	300.00	300.00	0.00	100.00%
Tithe	1,700.00	1,700.00	0.00	100.00%
Withdraw	600.00	600.00	0.00	100.00%
Gas	190.00	187.39	2.61	98.63%
Groceries	1,800.00	1,754.52	45.48	97.47%
Tammy's Nails	100.00	95.80	4.20	95.80%
Energy	220.00	208.88	11.12	94.95%
Gasoline	500.00	458.47	41.53	91.69%
Restaurant	500.00	403.64	96.36	80.73%
Water	250.00	198.30	51.70	79.32%
Total	**16,287.55**	**16,077.51**	**210.04**	**98.71%**

$16,287.55
Total Target

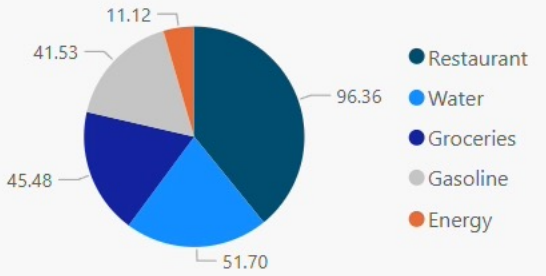
($42.96)
Negative Sum

$253.00
Positive Sum

Extra Revenue & Expense

Category	Sum of Amount
Auto Maintenance	290.24
Extra Budget	343.69
Money-Back Reward	60.19
Taxes and fees	25.87
Total	719.99

Top 5 Positive Balance



Exceeded Over Month



Month: April

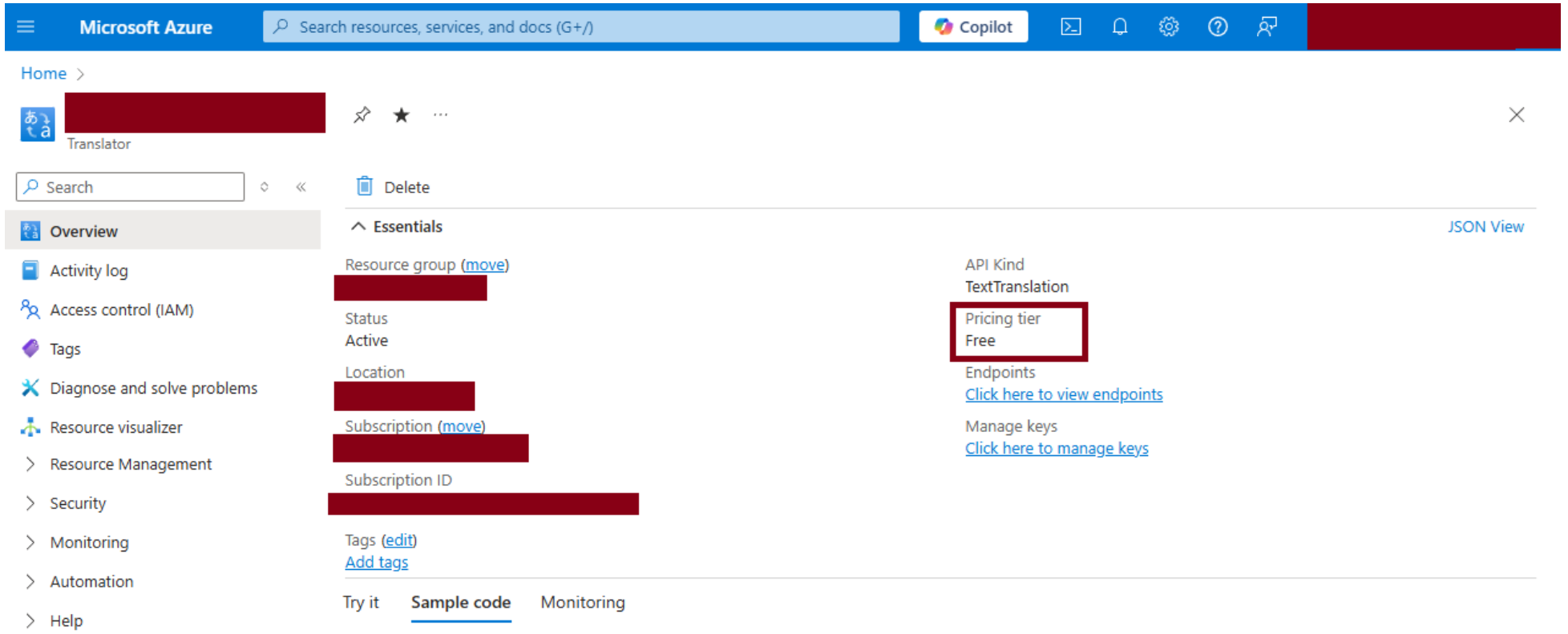
Year: 2025

Last Refresh
4/28/2025 3:22:05 PM

Main Flow | Trending | In x out | Utilities | Income Overview

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Azure AI Translator



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How to use:

- Endpoint
- Key
- From language
- To language
- Body

Try it | Sample code | Monitoring

Translate text instantly or in batches across more than 100 languages, powered by the latest innovations in machine translation. For more information about the sample code you can view our GitHub. Learn more

C# | Go | Java | Node.js | Python

Copy sample code | Open REST API guide | View on Github

☐ Insert my keys and region in sample code.

```python
import requests, uuid, json

# Add your key and endpoint
key = "<your-translator-key>"
endpoint = "https://api.cognitive.microsofttranslator.com"

# location, also known as region.
# required if you're using a multi-service or regional (not global) resource. It can be found in the Azure portal o
location = "<YOUR-RESOURCE-LOCATION>"

path = '/translate'
constructed_url = endpoint + path

params = {
    'api-version': '3.0'
    'from': 'en'
    'to': ['fr', 'zu']
}

headers = {
    'Ocp-Apim-Subscription-Key': key
    # location required if you're using a multi-service or regional (not global) resource.
    'Ocp-Apim-Subscription-Region': location
    'Content-type': 'application/json'
    'X-ClientTraceId': str(uuid.uuid4())
}
```




```python
df_category = spark.sql("SELECT * FROM my_lakehouse.category")

# 1. Get all unique values from a table
categories = df_category.select("Category").distinct().dropna().collect()

# 2. Translate row by row
translated_zh = []
for row in categories:
    original = row["Category"]
    try:
        result = client.translate(
            body=[{"text": original}],
            from_language="en",
            to_language=["zh"]
        )
        translated_zh.append((original, result[0].translations[0].text))
    except Exception as e:
        translated_zh.append((original, f"ERROR: {str(e)}"))
```

✓ 9 sec - Command executed in 9 sec 688 ms by Samantha Cruz on 8:13:00 AM, 6/14/25

Table | + New chart

Table view

	ABC Category	ABC Category_es	ABC Category_zh
1	Miscellaneous	Misceláneo	杂项
2	Savings Flexible	Ahorro Flexible	灵活储蓄
3	Water	Agua	水
4	Kids Expenses	Gastos de los niños	儿童消费
5	Gasoline	Gasolina	汽油
6	Car Insurance	Seguro de auto	汽车保险
7	Rent	Alquilar	租金
8	Energy	Energía	能源
9	Sports	Deportivo	体育
10	Money-Back ...	Recompensa de devolución de dinero	退款奖励
11	Extra Revenue	Ingresos adicionales	额外收入
12	Emergency Fu...	Fondos de Emergencia	应急资金
13	Groceries	Comestibles	杂货
14	Withdraw	Retirar	撤回
15	Auto Mainten...	Mantenimiento de automóviles	汽车保养
16	Internet	Internet	互联网
17	John Wage	John Wage	约翰·瓦格

Unpivot table

```python
from pyspark.sql.functions import lit

df = spark.sql("SELECT * FROM my_lakehouse.category")

langs = [("Category", "en"), ("Category_es", "es"), ("Category_zh", "zh")]

df_unpivoted = None
for col_name, lang in langs:
    temp = df.selectExpr(f"{col_name} as Category").withColumn("Language", lit(lang))
    df_unpivoted = temp if df_unpivoted is None else df_unpivoted.unionByName(temp)

display(df_unpivoted)
```

✓ 1 sec - Command executed in 1 sec 584 ms by Samantha Cruz on 9:27:14 AM, 6/14/25

Table | + New chart

Table view

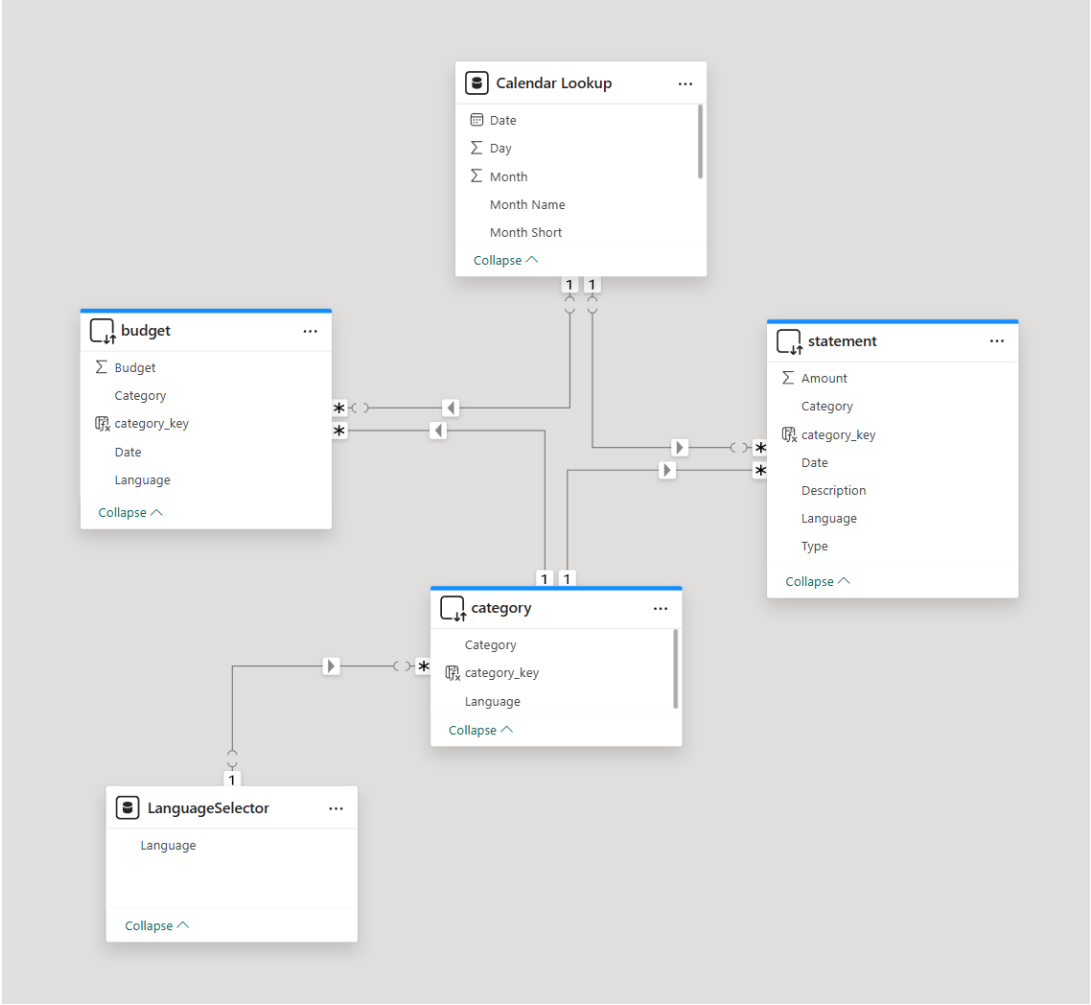
	ABC Category	ABC Language
59	Plan Móvil	es
60	Gas	es
61	Inversión	es
62	Netflix (en ing...	es
63	Oficina 365	es
64	Diezmo	es
65	杂项	zh
66	灵活储蓄	zh
67	水	zh
68	儿童消费	zh
69	汽油	zh

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Language Dimension

- A user picks a language from the slicer
- Power BI filters the category table by language
- That filtered translation is shown in visuals (matrix, tables, etc.)
- It works without switching pages or duplicating measures

Scales well with new languages (just add rows)
Avoids dynamic column/measure workarounds



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Monthly Expense Control

Category	Sum of Budget	Sum of Amount	Balance	% Spent
Miscellaneous	500.00	€ 534.78	-34.78	106.96%
Kids Expenses	350.00	€ 356.31	-6.31	101.80%
Woody the Dog	350.00	€ 351.86	-1.86	100.53%
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Donation	1,700.00	€ 1,700.00	0.00	100.00%
Emergency Funds	850.00	€ 850.00	0.00	100.00%
Internet	150.00	€ 150.00	0.00	100.00%
Investment	1,200.00	€ 1,200.00	0.00	100.00%
Mobile Plan	200.00	€ 200.00	0.00	100.00%
Netflix	30.45	€ 30.45	0.00	100.00%
Office 365	25.60	€ 25.60	0.00	100.00%
Rent	3,200.00	€ 3,200.00	0.00	100.00%
Savings Flexible	500.00	€ 500.00	0.00	100.00%
Sports	300.00	€ 300.00	0.00	100.00%
Tithe	1,700.00	€ 1,700.00	0.00	100.00%
Withdraw	600.00	€ 600.00	0.00	100.00%
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Groceries	1,800.00	€ 1,754.52	45.48	97.47%
Tammy's Nails	100.00	€ 95.80	4.20	95.80%
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Restaurant	500.00	€ 403.64	96.36	80.73%
Total	**16,287.55**	**€ 16,077.51**	**210.04**	**98.71%**

$16,287.55
Total Target

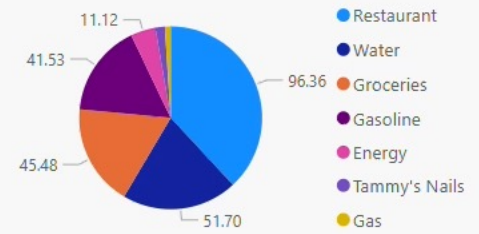
($42.96)
Negative Sum

$253.00
Positive Sum

Extra Revenue & Expense

Category	Sum of Amount
Auto Maintenance	€ 290.24
Extra Budget	€ 343.69
Money-Back Reward	€ 60.19
Taxes and fees	€ 25.87
Total	**€ 719.99**

Top Positive Balance



Exceeded Over Month



Language: en | es | zh (es)

Month: April

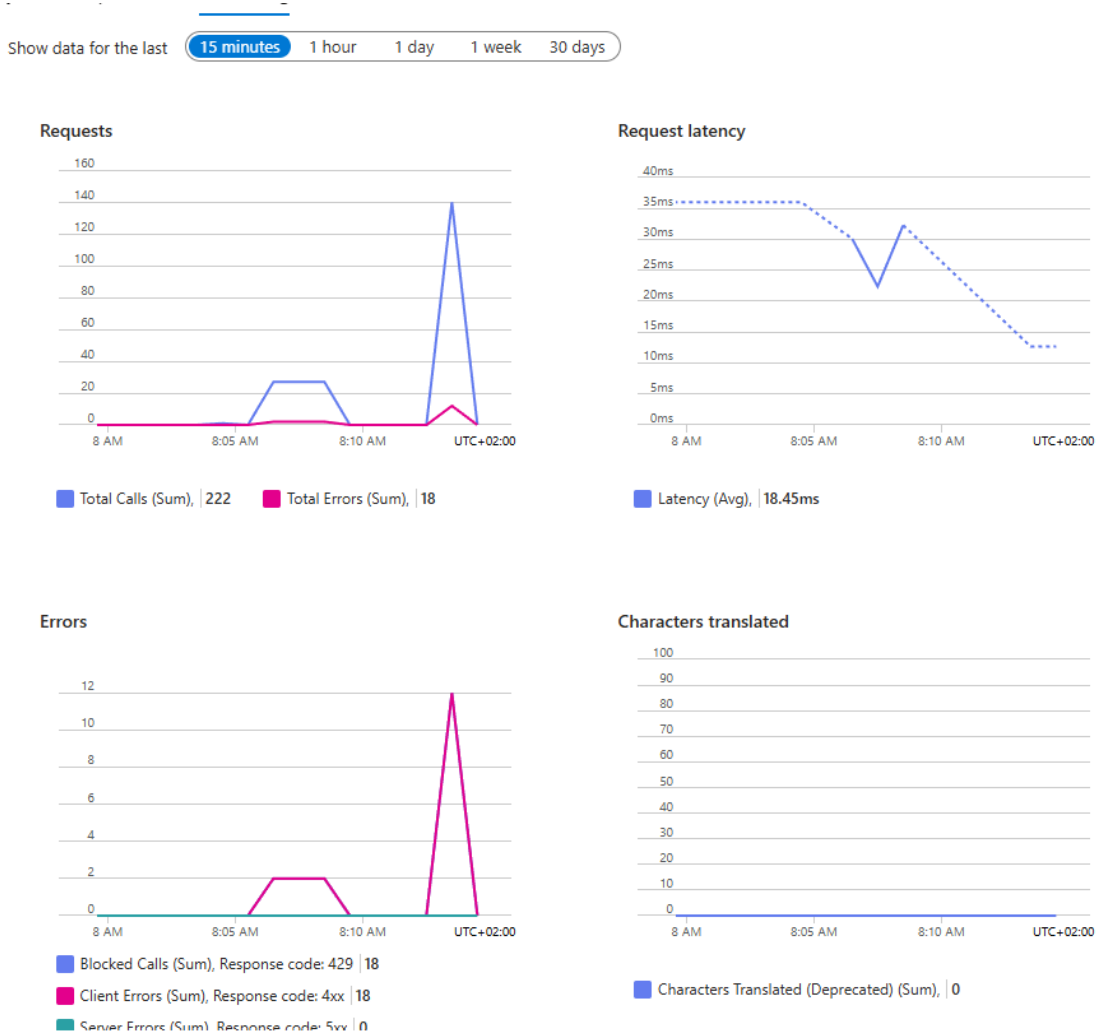
Year: 2025

Last Refresh: 6/14/2025 9:36:42 AM

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Monitoring

- Total Request Volume and Errors
- Latency and Performance Monitoring
- Client-Side Error Analysis (4xx)
- Monitoring Best Practices and Recommendations



Best Practices

1. You save if you send larger texts

- Azure AI Translator is billed by **number of characters sent**, but:
- Each request has **network and processing overhead**.
- Sending **multiple words in one request** is more efficient than one request per word.
- **Batch translation is supported**: up to **100 items per request**, and up to **5,000 characters per item**.

Best Practices

2. Alternatives to translating line by line:

- Group unique text values
- Batch translate multiple entries
- Concatenate short texts into one string

```
body = [{"text": val} for val in values]
response = client.translate(
    body=body,
    from_language="en",
    to_language=["es"]
)
```

Limitations

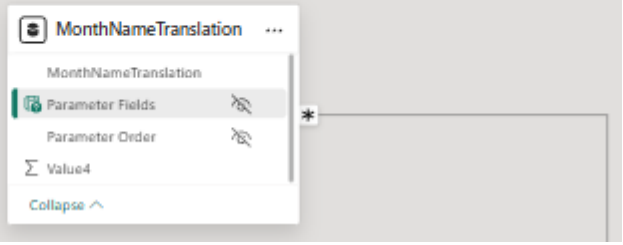
- Semantic Link Labs works only with semantic model metadata
- Azure AI Translator helps translate text, but does not apply it automatically
- Requires custom strategies to expose and manage translations
- Report layer elements (e.g. titles, buttons, text boxes) are not accessible via APIs
- Full automation of multilingual reports is not yet supported end-to-end
- Manual or embedded-layer solutions are needed for visual-level translations

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Parameterized Translation Tables

- Central **DimLanguage** table as language slicer
- Individual translation tables for each entity
- Composite key: [Entity ID] + [Language Code]
- Translated value stored in a Value or Valued column
- Relationships drive context filtering based on language

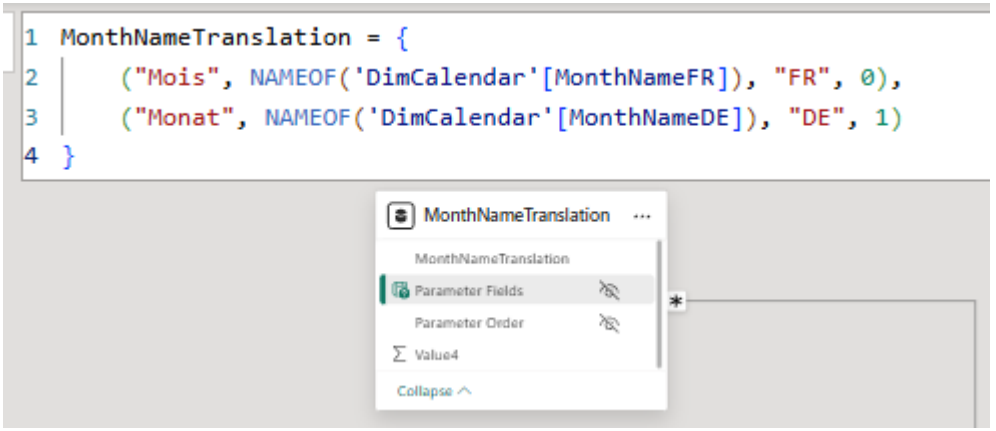
```
1 MonthNameTranslation = {
2     ("Mois", NAMEOF('DimCalendar'[MonthNameFR]), "FR", 0),
3     ("Monat", NAMEOF('DimCalendar'[MonthNameDE]), "DE", 1)
4 }
```



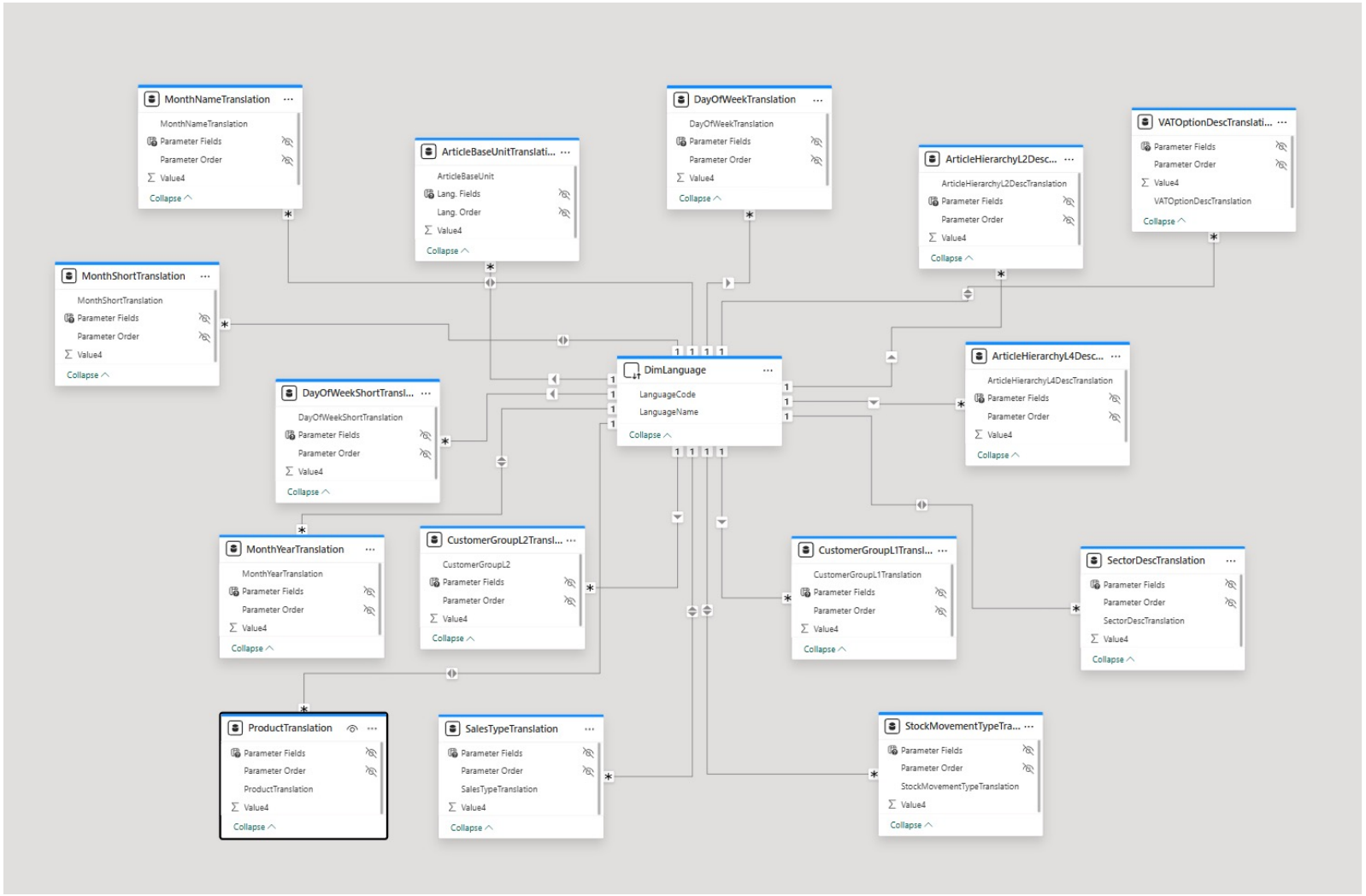
Parameterized Translation Tables

Benefits

- Centralized control of translations
- Scales across multiple entities and visuals
- Fully dynamic – responds to user language selection
- Clean separation between data and presentation layer
- Supports automation and governance



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